NLP Project 14

UNIVERSITY OF OULU

Mining nature from the Web

How does a search engine see ‘climate change’ ?

521158S • Natural Language Processing

Van-Anh Diep Lison Bernard Vladislav Ekushev

27.10.2026

Introduction

Our project aims to evaluate how the search engine can teach the community by promoting documents that can deliver positive messages about a target word

and what other patterns show up?

Pages ranked higher by Google deliver a more positive message about climate change than pages further down

If so, people who stop at the first page see the hopeful side first.

Top results raise more awareness of causes, impacts and solutions, and cite better sources

Search engines say they reward authority. We check whether an LLM reading the pages agrees.

Process

1. Collect the snippets
2. Analyse the snippets
3. Read pages with an LLM
4. Compare by rank

Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank

```python
# Google via SerpAPI, US
params = {
  "engine": "google",
  "q": "climate change",
  "start": page * 10,
  "hl": "en", "gl": "us",
}

# keep only URLs not seen yet
new = [r for r in items
       if r.url not in seen]

# stop at 100 unique results
if len(results) ⩾ 100: break
```

1. Query “climate change” on Google via SerpAPI
2. Keep the first 100 unique URLs
3. Save rank, title, URL and snippet
4. Cache results

```python
@dataclass
class SearchResult:
    rank: int
    title: str
    url: str
    snippet: str

with pd.ExcelWriter(out) as xw:
    pd.DataFrame(map(asdict, results))
      .to_excel(xw, "snippets",
index=False)
    pd.DataFrame(meta.items())
      .to_excel(xw, "meta", index=False)

# Paid SerpAPI request
uv run collect -p serpapi

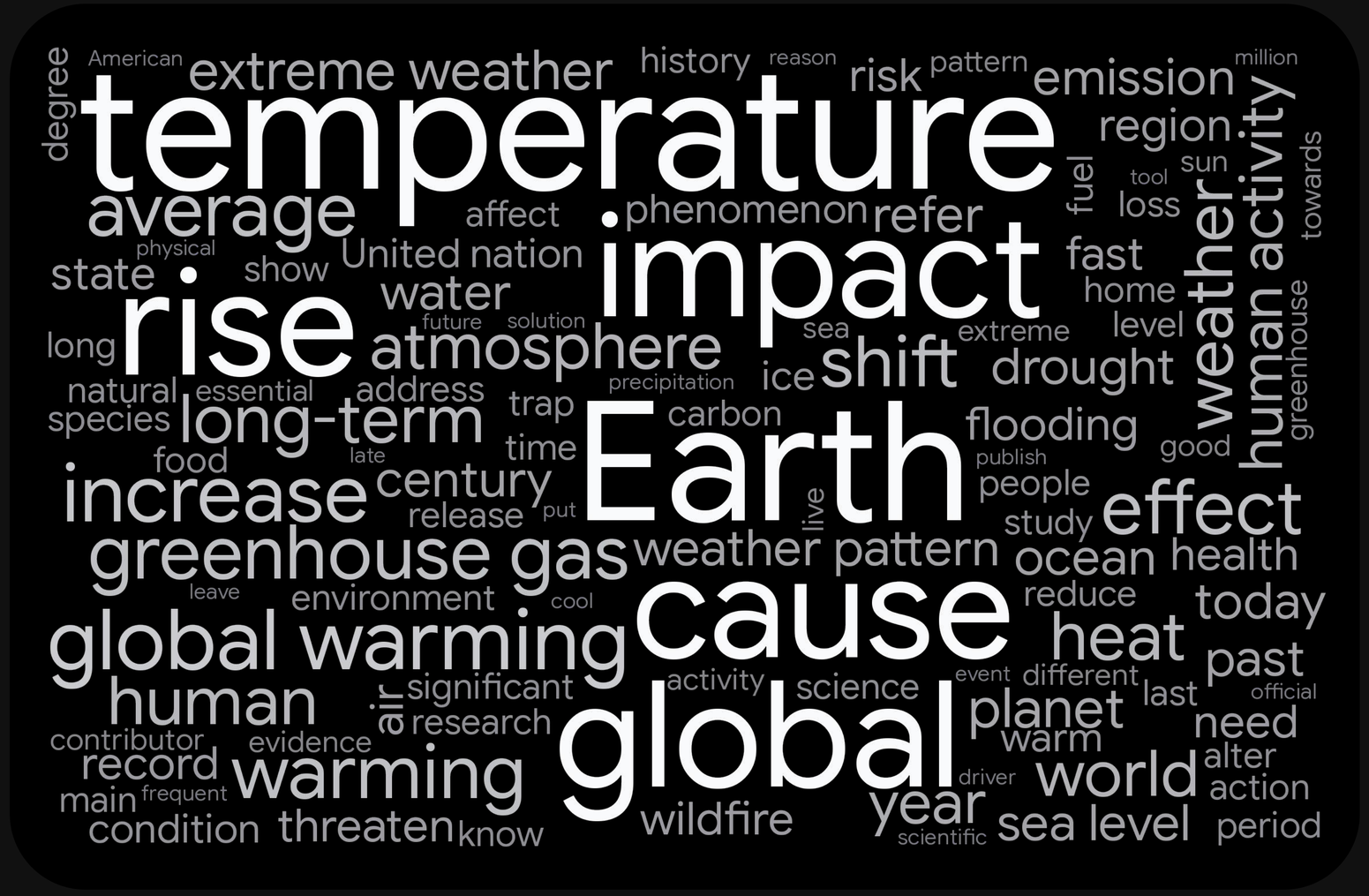
# Free ddgs library
uv run collect -p ddgs -e yahoo -r us-en
```

Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank



What the snippets talk about

Words after cleaning

2,209

Snippets explain what climate change is, rarely what to do about it.

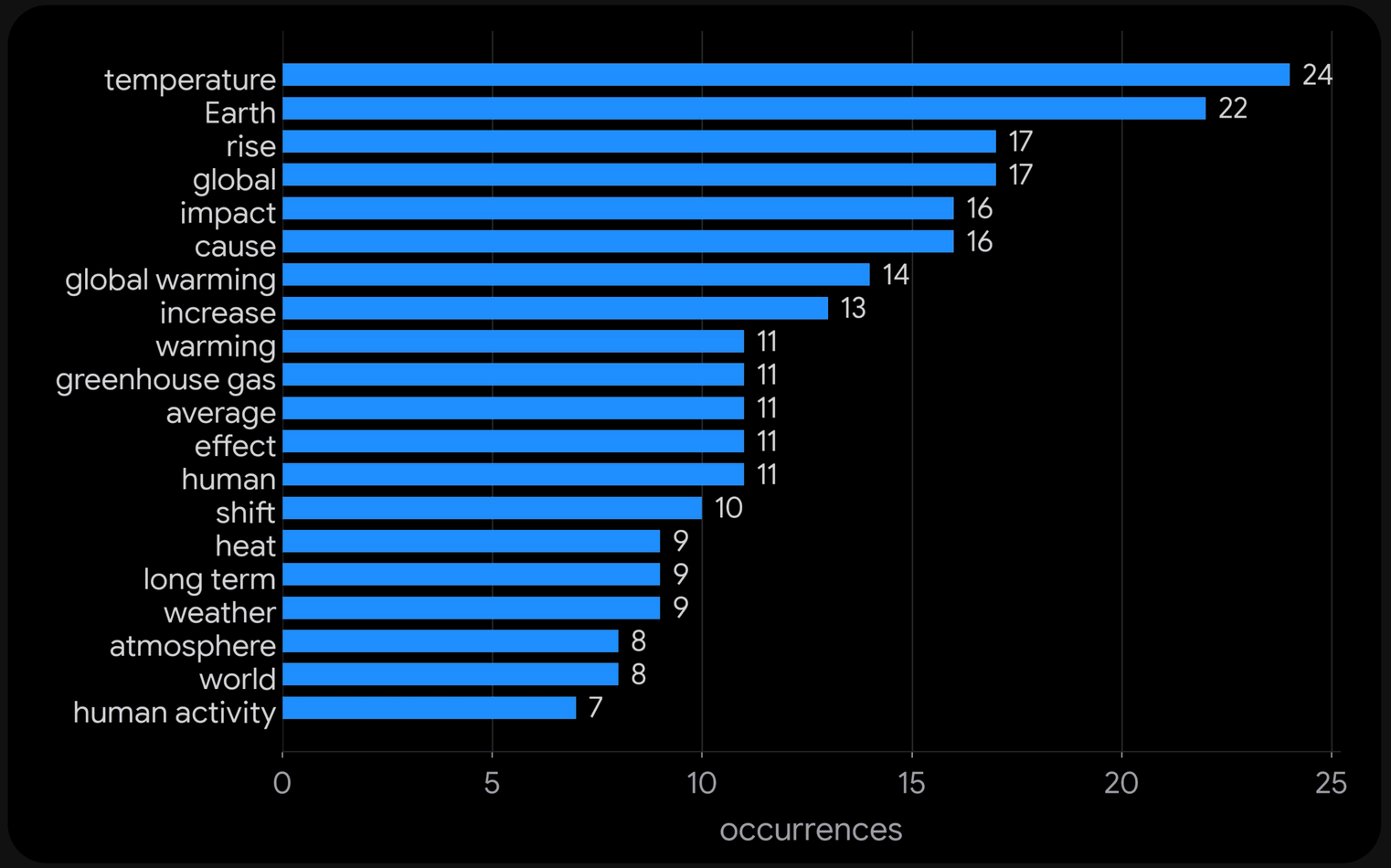
Without ‘climate’ and ‘change’, physical and causal words lead and impacts come second. ‘Solution’, ‘reduce’ and ‘address’ barely appear.

Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank



Most frequent terms

Top term, 100 snippets

temperature · 24

Counts after cleaning and lemmatising, with “climate” and “change” removed.

Two-word collocations count as one term: global warming, greenhouse gas, long term, human activity.

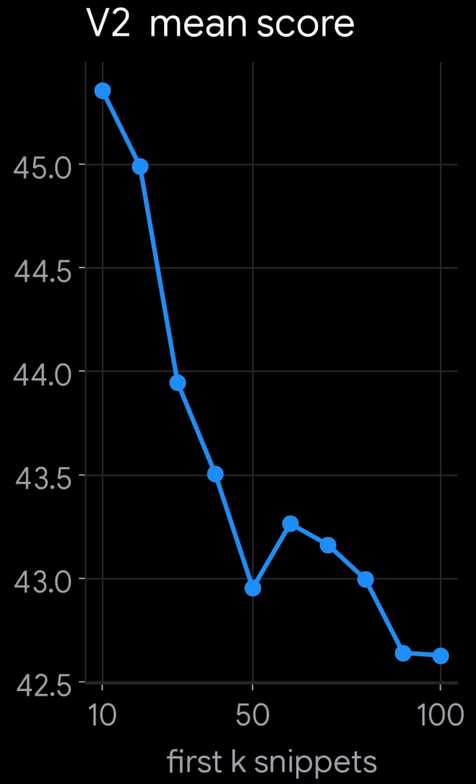
Collect the snippets

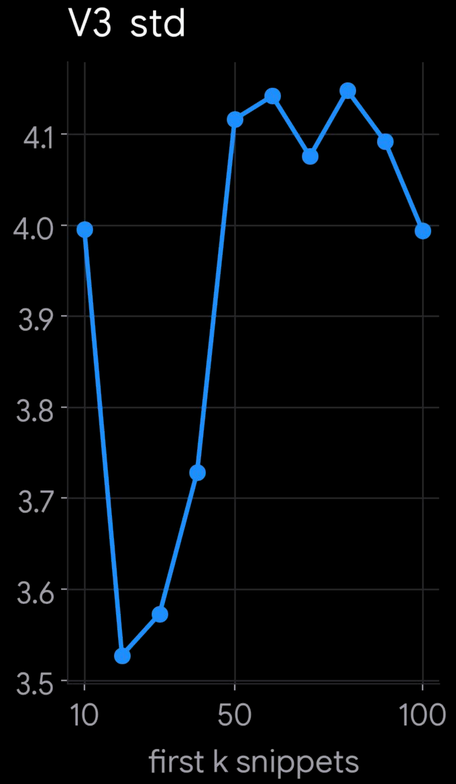
Analyse the snippets

Read pages with an LLM

Compare by rank







Do the results look alike?

Identical pairs (V1)

0

Mean score (V2)

45.4 → 42.6

FuzzyWuzzy score on every pair

k(k−1)/2 pairs

V1 = identical pairs
V2 = mean score
V3 = standard deviation, for k = 10, 20 ... 100.

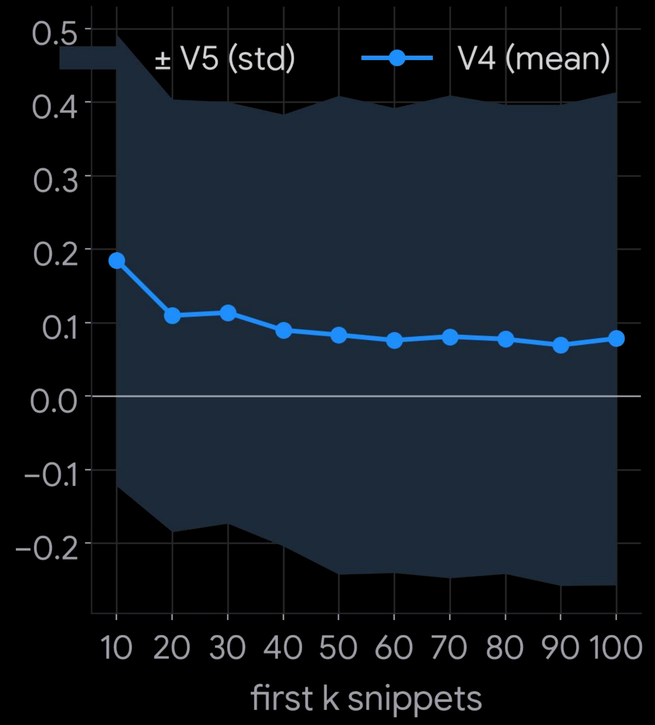
score 0–100, similarity of characters

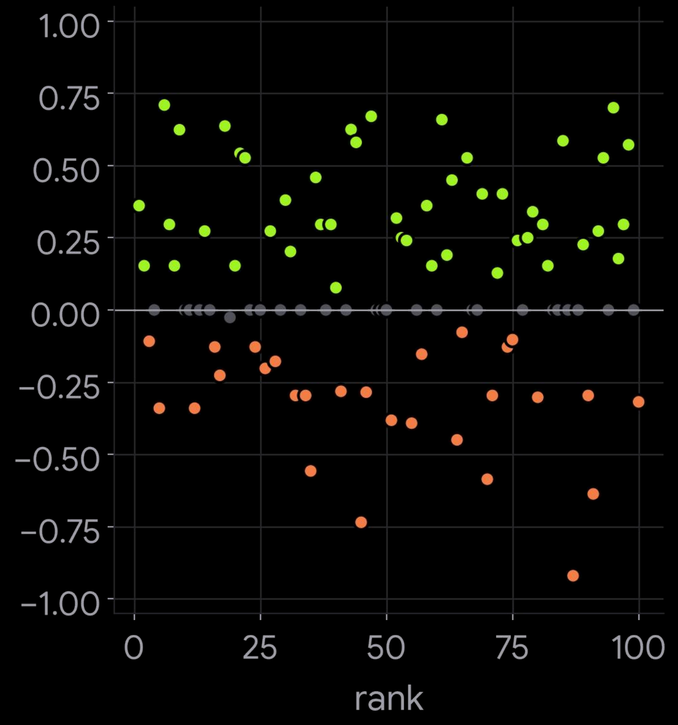
Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank





V4 = mean
V5 = standard deviation over the first k

46 positive
28 negative
26 neutral

How positive are they?

Mean VADER, top 10

+0.185

Mean VADER, all 100

+0.078

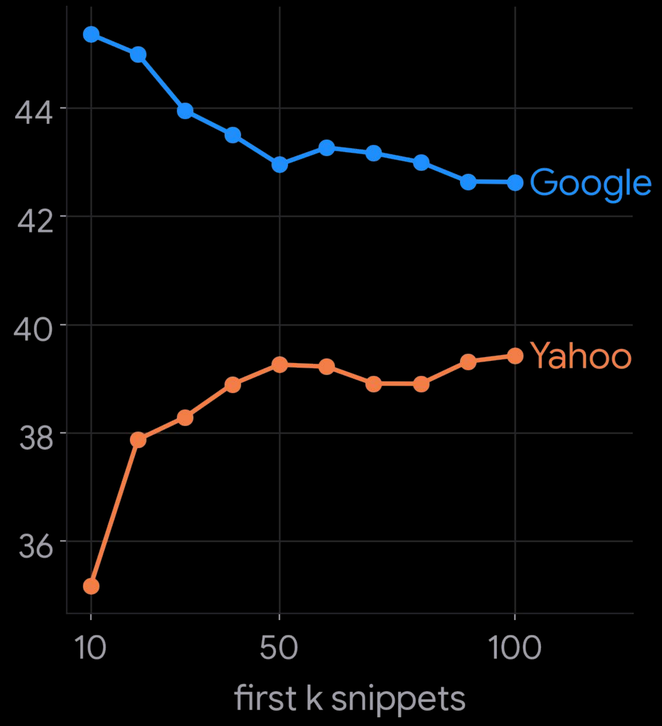
Close to neutral, slightly positive

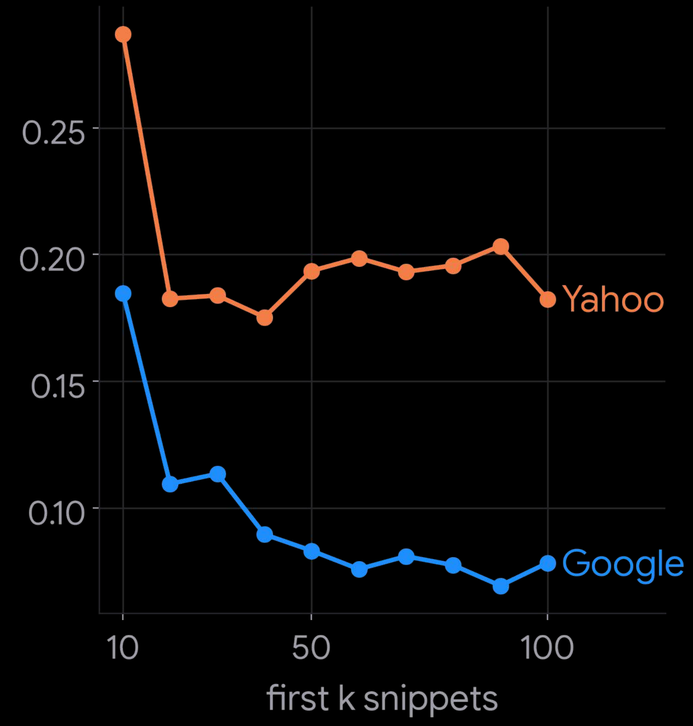
Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank





Google vs Yahoo

Shared top-10 domains

7 / 10

Shared URLs

37 / 100

Yahoo's snippets are less repetitive and more positive.

Logged-in and per-country runs are not analysed yet.

Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank

From the point of view of an LLM

URL

↓

Page text

↓

Gemini API

↓

Fixed JSON

JSON fields

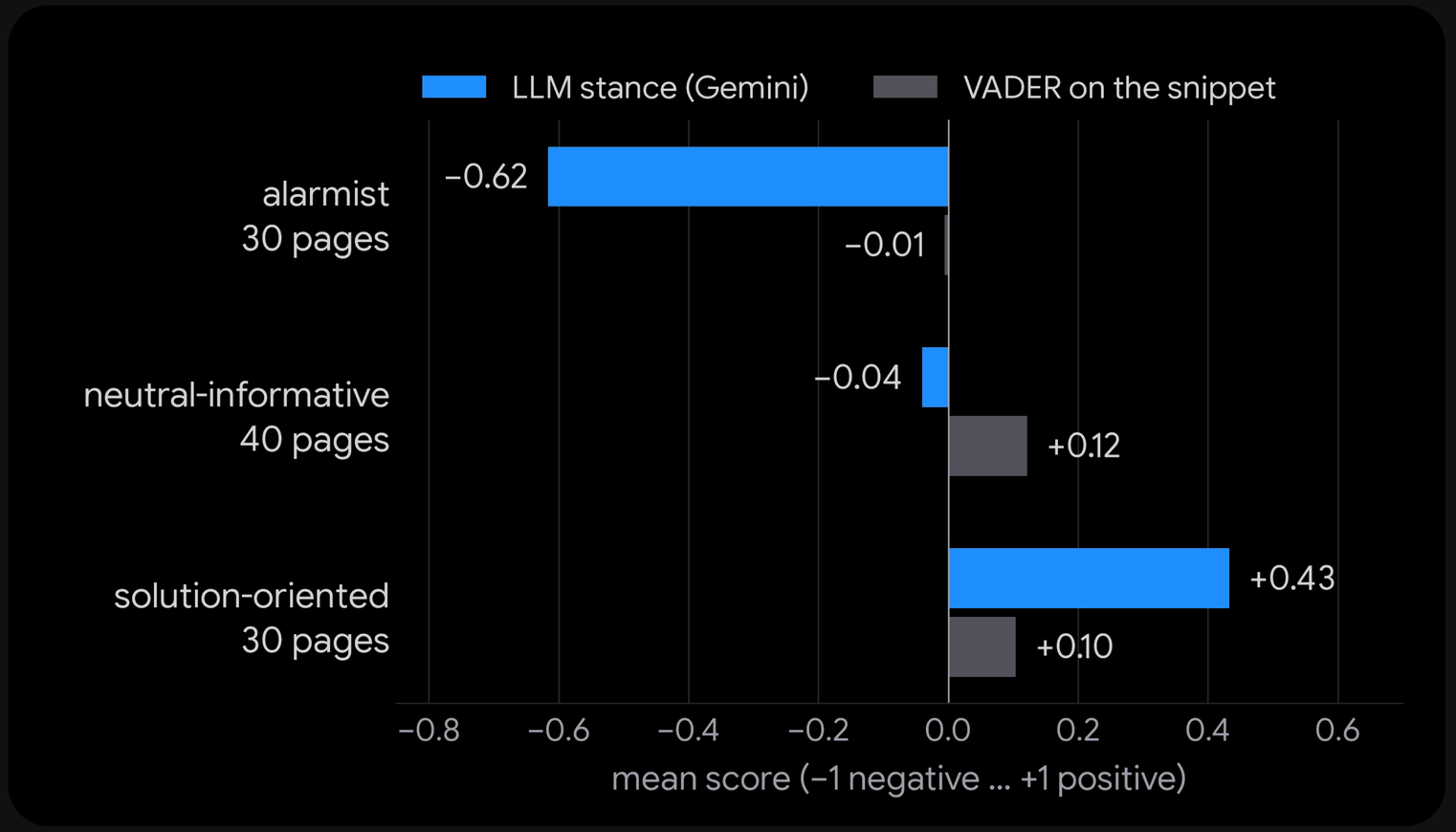
```
Collect the snippets Gemini's answer for
rank 1, un.org
takeaway: Fossil fuels drive warming
actions: Switch to renewable,...
stance: 0.4
stance_label: solution-oriented
awareness: 5
references: UN reports, UNFCCC
reference_quality: 4
reference_rationale: UN reports ..
```

Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank



Lexicon vs LLM

Alarmist pages, LLM

−0.62

Alarmist pages, VADER

−0.01

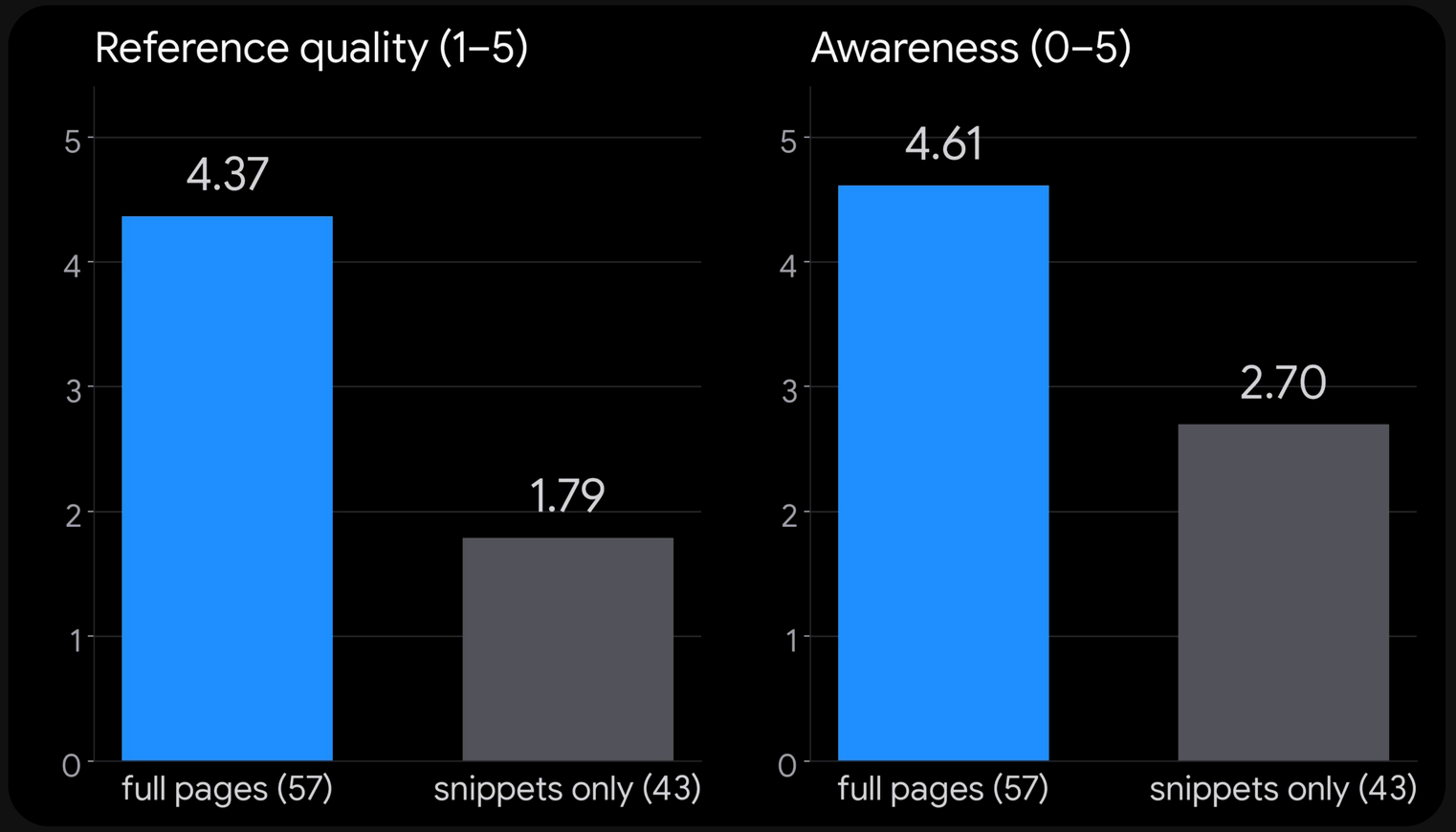
VADER only sees the words of a short snippet
The LLM reads the page.

Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank



Full pages vs snippets

Full pages / snippets only

57 / 43

Neutral / solution / alarmist

40 / 30 / 30

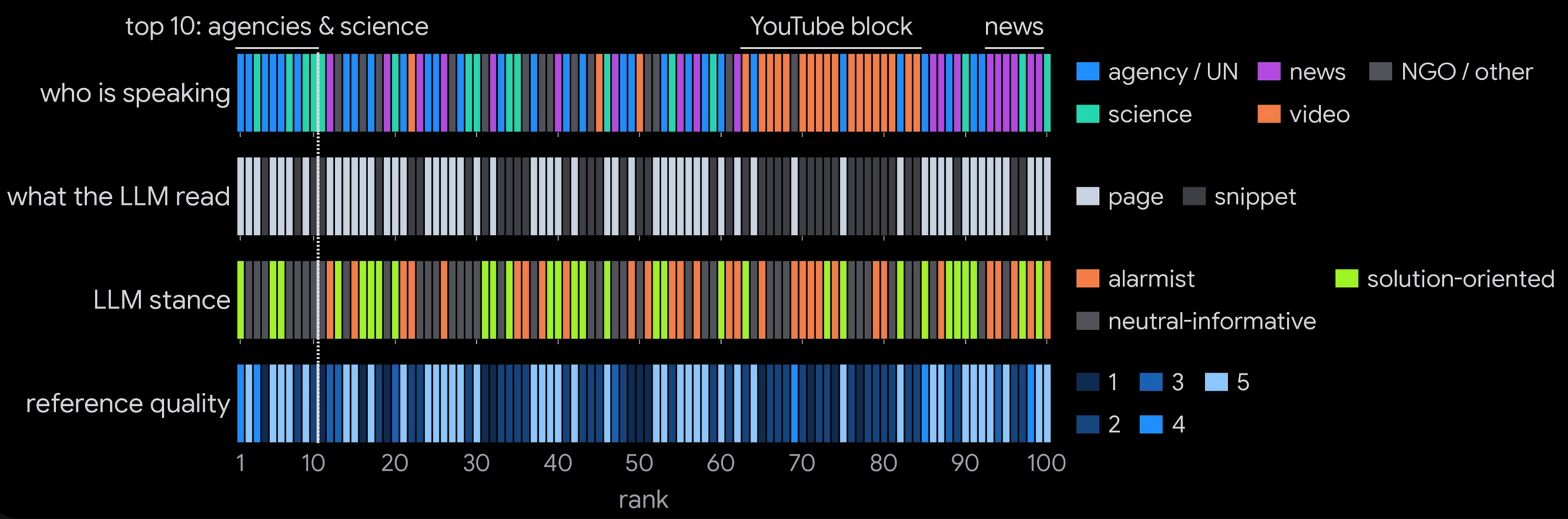
On full pages the LLM rates references 4.4 and awareness 4.6. Snippet-only results score lower.

Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank



Half the pages are alarmist

ranks 88-100

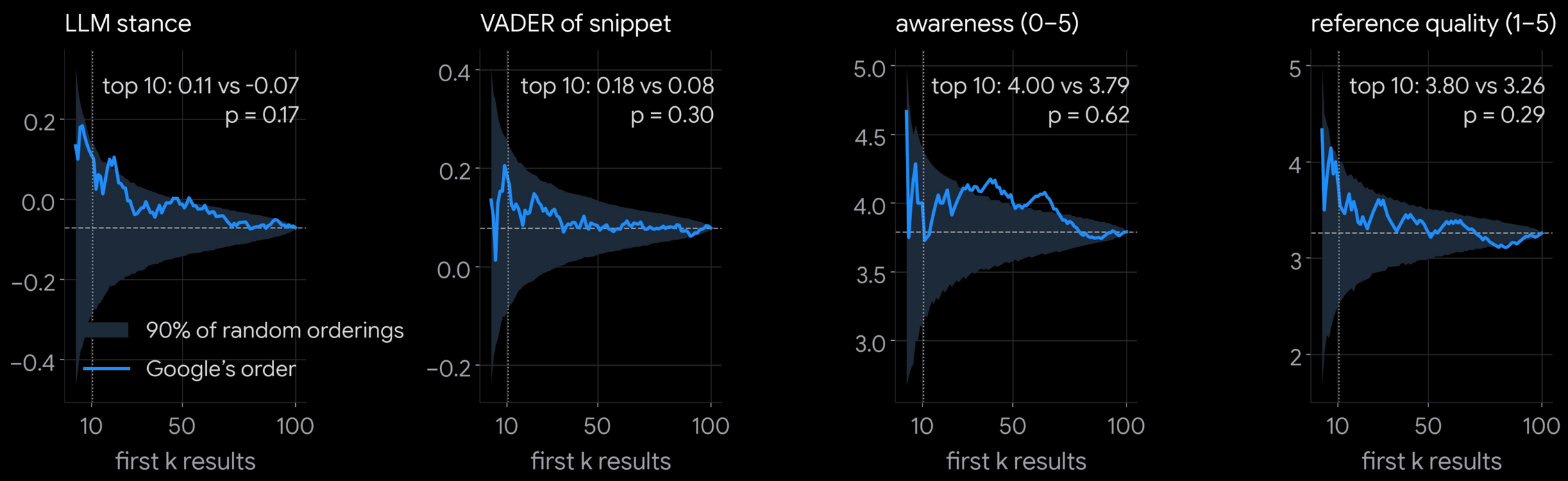
Mostly news

Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank



Top-10 stance

0.11 / −0.07

p-value

0.17

does rank order matter?

Not measurably. For every metric, Google's running mean stays inside the band of 2,000 random orderings.

Collect the snippets

Analyse the snippets

Read pages with an LLM

Compare by rank



Fewer pages mention solutions and impacts past rank 50

Full pages only, ranks 1–50 vs 51–100.
Fisher's exact test: p = 0.035 and p = 0.006.

Mention solutions

71% → 42%

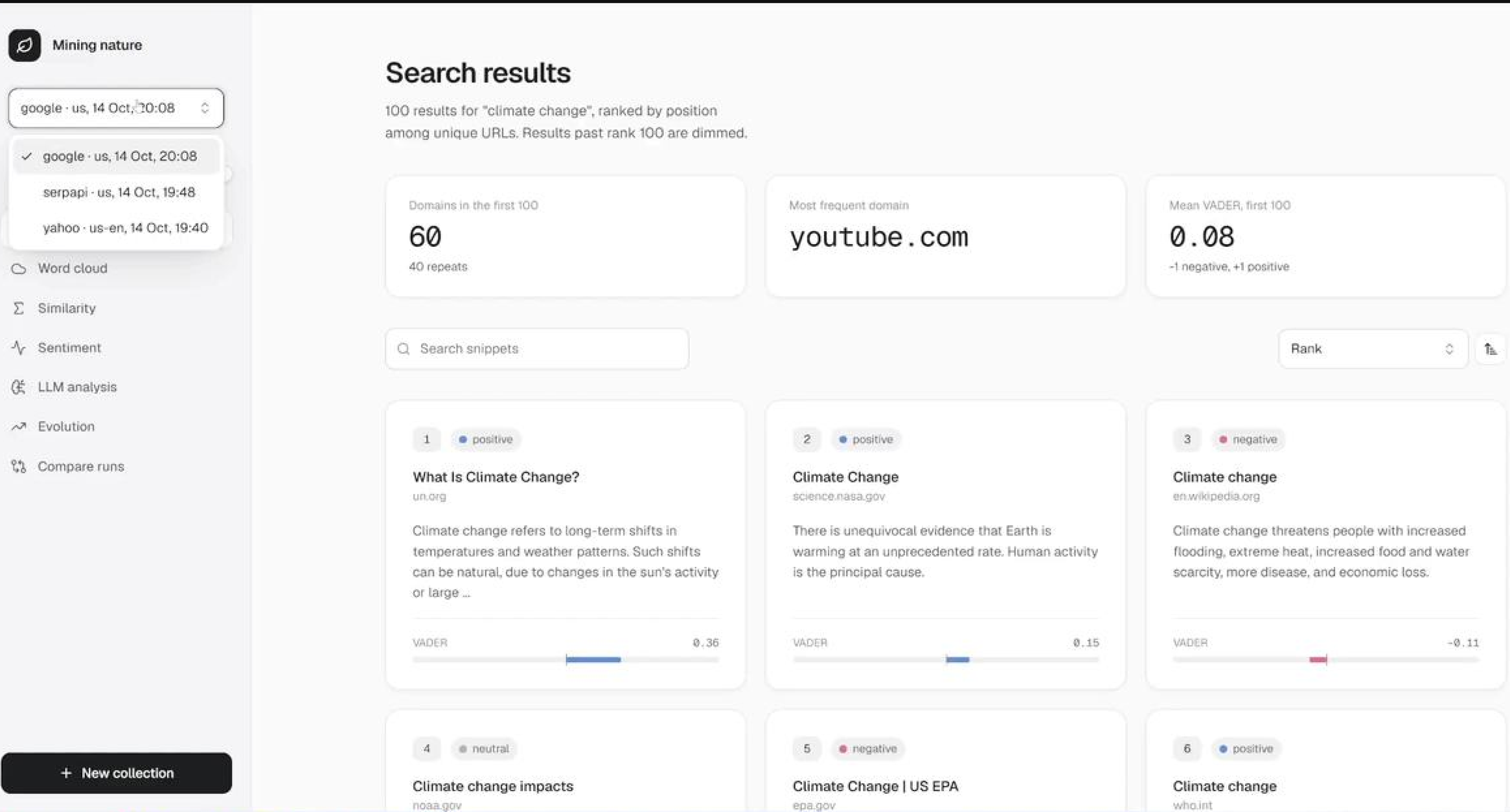
Mention impacts

90% → 58%

Mentions of causes stay about the same.

GUI

React web app + FastAPI backend



Conclusion

Google's top results explain climate change. They are neutral rather than positive.

Hypotheses held

0 / 2



The top 10 is agency and science explainers: 7 neutral, 3 solution-oriented, 0 alarmist

Rank order barely predicts tone, awareness or source quality: no top-10 difference is larger than random ordering produces

After the rank 50, fewer pages mention solutions (71% → 42%) and impacts (90% → 58%)